

"The highest spike of 7,466 new COVID-19 cases and 175 deaths in the last 24 hours has been reported in the country. The total number of corona virus infected in the country now stands at 165,799 including 89,987 active cases, 71,105 cured/discharged/migrated. ...

...Meanwhile, a sharp rise in coronavirus has been reported in Mumbai, Maharashtra, Delhi, West Bengal and Ahmedabad. Haryana has ordered sealing of its borders with the national capital."

```
import matplotlib.pyplot as pPlot
from wordcloud import WordCloud,
STOPWORDS
import numpy as npy
from PIL import Image
text = """Highest spike of 7,466 new
COVID-19 cases and 175 deaths in the
last 24 hours has been reported
in the country. Total number of
coronavirus infected in the country now
stands at 1,65,799 including 89,987
active cases, 71,105 cured/discharged/
migrated.
```

```
Meanwhile, a sharp rise in coronavirus
has been reported in Mumbai,
Maharashtra, Delhi, West Bengal,
Ahmedabad. Haryana has ordered sealing
of its borders with the national
capital."""
```

```
cloud = WordCloud(background_color =
"green", max_words = 100, stopwords =
set(STOPWORDS))
cloud.generate(text.lower())
cloud.to_file("Fig3.png")
```

OUTPUT

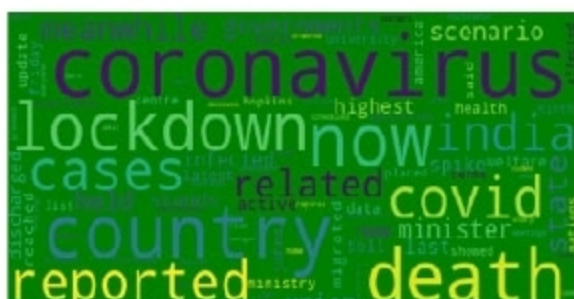


Figure 3: Word Cloud

The future of NLP

NLP provides a wide range of tools and techniques that can be applied to all areas of life. By learning them and using them in our day to day dealings, the quality of life can greatly improve. Companies from high tech startups to multinationals see the use of NLP as a key competitive advantage in an increasingly competitive and technology-driven market.

AI virtual assistants can be made smarter, thereby extending NLP experiences. An AI medical assistant will understand conversations using NLU models enabled with medical terminology. Trained in medical terminology and data, it would be able to listen and interpret conversations between a doctor and a patient to record and subsequently recap the conversation as notes. The frequent exposure of healthcare staff to dangerously infected patients can be avoided. Today we see the novel coronavirus spreading across the globe, threatening the lives of healthcare staff. Using NLP trained virtual assistants, we can reduce these risks and control the pandemic. Treatment becomes convenient and faster. Lower use of medical kits (like masks, gloves, etc) will in turn be cost-effective and environment-friendly.

As the NLP technology evolves, computer systems will increase their current capabilities from processing to figuring out human language in a holistic way. Until now, NLP has only been able

to infer a limited variety of human emotions like joy or anger. Eventually, it will be programmed to recognise more complicated aspects of human language, which includes humour, parody, sarcasm, satire, irony and cynicism

Natural language processing is set to be integrated in almost every aspect of life, so the sky is the limit. The future is going to see some massive changes as the technology becomes more mainstream and advanced. Another major consequence of artificial intelligence and natural language processing will be the invasion of robots in the workplace; so industries everywhere have to start preparing for this.

Some of us remember 'Wall-E', the well-known sci-fi film that was released in 2008 in which a solitary trash compactor robot spends his days changing garbage into string blocks and building structures with them. One day, a giant spaceship arrives and drops a new robot named Eve on Earth. Eve has been despatched to our planet to determine whether life exists here! Eve focuses on finding the existence of life, ignoring Wall-E entirely as he is a non-living thing. However, Wall-E rescues Eve from a dust storm and provides her with shelter, where he shows her the last living plant. By this stage in the movie, Wall-E has developed a personality, is lonely, and falls in love with Eve.

Could this become a reality in the near future? Are we heading in the direction of giving birth to fully conscious machines?

With Cortana, Alexa and Siri, the next-gen NLP journey has begun. **END** 🐧

By: Dibyendu Banerjee

The author is a senior architect at Cognizant's AI & Analytics. practice. He has more than 14 years' experience on an optimal blend of domain expertise, diverse technology competence, and project management skills, which includes Python, R, Java, and open source technologies. He shares the latest trends around cutting-edge technologies through blogs, white papers, etc.